```
    for point in node.points
do                                                  #4
      if region.intersectsPoint(point)
then                                          #5

points.insert(point)
       #6
  else
   for child in node.children
do                                                 #7
      if region.intersectsNode(child) then
                           #8
        points.insertAll(pointsWithinRegion(child,
region))                   #9
  return
points
      #10
```

#1 Function `pointsWithinRegion` takes a node, on which the search must be performed, and the search region. It returns a list of points stored in the subtree rooted at `node` and lying within the search region (in other words, the intersection between the region and the node's bounding envelope).
#2 Initializes the return value to an empty list.
#3 Check if the node is a leaf.
#4 Cycles through all points in current leaf.
#5/#6 If current point is within the search region, adds it to the list of results. The onus of providing the right method to check if a point lies in a region is on the region's class (so regions of different shape can implement the method differently).
#7 If, instead, the node is internal, cycle through its children.